Jing-Zhe Xu

Curriculum Vitae

Assistant Professor | Hunan University

Department of Control Science and Engineering
School of Artificial Intelligence and Robotics
Hunan University, Changsha, China

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Academic Appointment

Assistant Professor, Hunan University | 2026 - present

School of Artificial Intelligence and Robotics, Department of Control Science and Engineering.
Core member, National Engineering Research Center for Robot Visual Perception and Control Technology.

Education

PhD in Control Science and Engineering | September 2022 - June 2026

Huazhong University of Science and Technology
Supervisors: D.-X. He and Z.-W. Liu; co-supervisor: M. Chi.

Master's Degree in Mechanical Engineering | September 2019 - June 2022

China University of Geosciences (Wuhan)
Supervisor: M.-F. Ge.

Bachelor's Degree in Mechanical Design, Manufacturing and Automation

China University of Geosciences (Wuhan) | September 2015 - June 2019

Research Interests

Distributed optimization, game-theoretic decision-making and cooperative control for networked systems and multi-robot systems, with a focus on linking distributed decisions to physical dynamics and execution constraints.

Delay-tolerant distributed optimization

Distributed optimization and control of nonlinear networked systems with input delays and communication constraints.

Dynamic games and resilient reconfiguration

Rapid equilibrium seeking and task reconfiguration in open multi-agent systems with changing membership, constraints and node failures.

Human and AI in the loop

Multi-robot teleoperation and cooperative execution under limited communication and uncertain dynamics.

Selected Publications

Representative journal articles; author order follows the publications. * Corresponding author.

[1] **J.-Z. Xu**, Z.-W. Liu*, M.-F. Ge, Y.-W. Wang, and D. He, "Semiglobal input-delay tolerance algorithm for distributed nonconvex optimization of networked nonlinear system," IEEE Transactions on Automatic Control, 2026, doi: 10.1109/TAC.2026.3702930.

[2] **J.-Z. Xu**, Z.-W. Liu*, M.-F. Ge, T. Yang, M. Chi, and D. He, "Distributed predefined-time algorithms for optimal solution seeking in multi-agent systems subject to input disturbances," Automatica, vol. 174, Art. no. 112139, 2025.

[3] **J.-Z. Xu**, M.-F. Ge*, Z.-W. Liu, W.-Y. Zhang, and W. Wei, "Force-reflecting hierarchical approach for human-aided teleoperation of NRS with event-triggered local communication," IEEE Transactions on Industrial Electronics, vol. 69, no. 3, pp. 2843-2854, 2022. ESI Highly Cited Paper in 2022.

[4] **J.-Z. Xu**, M.-F. Ge*, Z.-W. Liu, H. Yan, C.-D. Liang, and Y.-D. Wu, "Hierarchical cloud-supported control framework for space teleoperation with networked-follower robots," IEEE/ASME Transactions on Mechatronics, vol. 29, no. 5, pp. 3300-3311, 2024.

[5] **J.-Z. Xu**, Z.-W. Liu*, M.-F. Ge, Y.-W. Wang, and D.-X. He, "Self-triggered MPC for teleoperation of networked mobile robotic system via high-order estimation," IEEE Transactions on Automation Science and Engineering, vol. 22, pp. 6037-6049, 2025.

[6] **J.-Z. Xu**, Z.-W. Liu, X. Sun, D. He, and M.-F. Ge, "Distributed Time-Varying Optimization for Predefined-Time Affine Formation in MASs," IEEE Transactions on Automation Science and Engineering, vol. 23, 2026, doi: 10.1109/TASE.2026.3734449.

[7] **J.-Z. Xu**, Z.-W. Liu*, D.-X. He, Z. Jia, and M.-F. Ge, "Dynamic Nash equilibrium seeking for constrained non-cooperative optimization of open multi-agent systems," IEEE Transactions on Systems, Man, and Cybernetics: Systems, vol. 55, no. 6, pp. 3846-3855, 2025.

[8] **J.-Z. Xu**, H.-L. Peng, M.-F. Ge*, Z.-W. Liu, H. Yan, and J. Liu, "Hierarchical prescribed-performance nonlinear control for teleoperation of networked secondary robots with coupling constraints," IEEE Transactions on Control of Network Systems, vol. 12, no. 1, pp. 119-131, 2025.

[9] **J.-Z. Xu**, Z.-W. Liu*, M.-F. Ge, Y.-W. Wang, and D. He, "Distributed robust predefined-time algorithm for seeking Nash equilibrium in MASs," IEEE/CAA Journal of Automatica Sinica, vol. 12, no. 5, pp. 1053-1055, 2025.

[10] **J.-Z. Xu**, Z.-W. Liu*, D.-X. He, M.-F. Ge, and M. Chi, "Robust predefined-time control for optimal formation of networked mobile vehicle systems," IEEE/CAA Journal of Automatica Sinica, vol. 12, no. 4, pp. 824-826, 2025.

Research Projects

Principal Investigator | January 2025 - December 2026

Optimization and game-theoretic control of hybrid networked systems in highly adversarial scenarios.
National Natural Science Foundation of China (NSFC), Basic Research Program for Young Students (Doctoral Students).

Project Participant | January 2024 - December 2027

Secure cooperative control of swarm intelligence systems in open and uncertain environments.
National Natural Science Foundation of China (NSFC), General Program.

Patents and Patent Applications

Granted Chinese invention patent

J.-Z. Xu et al. Neuro-adaptive fixed-time control method for complex teleoperation. CN111427264B.

Chinese invention patent applications

J.-Z. Xu et al. Device for automatic coolant replenishment and concentration maintenance. CN107562080A.

J. Dong, K. Huang, C.-D. Liang, M.-F. Ge, P. Su, J.-Z. Xu, et al. Finite-time cooperative tracking of multiple targets by multi-agent systems and a storage medium. CN113400299A.

P. Su, K. Huang, C.-D. Liang, M.-F. Ge, J. Dong, J.-Z. Xu, et al. Cooperative control of networked robots: method, apparatus, equipment and storage medium. CN113359439A.

X. Duan, C. Li, Y. Zhou, Y. Shuai, Y. Huang, J.-Z. Xu, et al. Automatic fixed-length feeding device based on a band saw. CN108213579A.

T.-F. Ding, B. Li, T. Zhou, M. Li, J. Zhao, J.-Z. Xu, et al. High-density intelligent three-dimensional storage system for steel. CN106809587A.

Honors and Awards

2025 | Outstanding Student Presentation, 21st Forum on Network Science and Engineering

2025 and 2024 | National Scholarship for Doctoral Students

2021 | National Scholarship for Master's Students

Academic Service

Journal reviewer: Automatica; IEEE Transactions on Vehicular Technology; IEEE/ASME Transactions on Mechatronics; IEEE Transactions on Industrial Informatics.